Vectors I Basics Worksheet: Questions

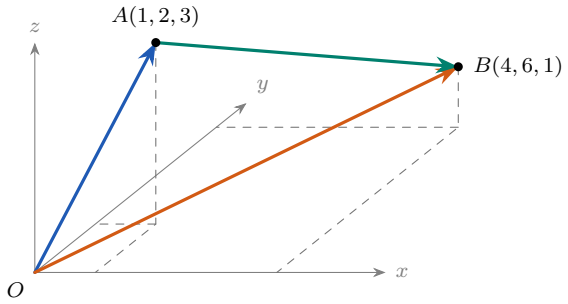
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Basics of vectors: **i**, **j**, **k** notation, magnitude, unit vectors, parallel vectors, collinear points, the ratio theorem, the scalar product and the vector product.

1 i, j, k Notation and Column Vectors

1 Write $\mathbf{a} = 3\mathbf{i} - 2\mathbf{j} + 5\mathbf{k}$ as a column vector, and write $\mathbf{b} = \begin{pmatrix} 4 \\ 0 \\ -1 \end{pmatrix}$ in terms of **i**, **j** and **k**.

2 The points A and B have coordinates $(1, 2, 3)$ and $(4, 6, 1)$. Find $\overrightarrow{AB}$ in terms of **i**, **j** and **k**.



3 **(Tricky)** Given $\mathbf{p} = 2\mathbf{i} + \mathbf{k}$ and $\mathbf{q} = -\mathbf{i} + 3\mathbf{j}$, find $\mathbf{r}$ such that $2\mathbf{p} - 3\mathbf{q} + \mathbf{r} = \mathbf{0}$. Give $\mathbf{r}$ in terms of **i**, **j** and **k**.

For the following statements, state whether it is true or false.

(a) $3\mathbf{i} + 2\mathbf{k} = \begin{pmatrix} 3 \\ 2 \\ 0 \end{pmatrix}$

(b) The **j**-component of $4\mathbf{i} - \mathbf{j} + 2\mathbf{k}$ is -1

(c) The position vector of $P(1, -2, 0)$ is $\mathbf{i} - 2\mathbf{j}$

(d) $\overrightarrow{AB} = \overrightarrow{OA} - \overrightarrow{OB}$

(e) $\mathbf{i} + \mathbf{j} + \mathbf{k} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$

2 Magnitude

1 Find $|2\mathbf{i} - 3\mathbf{j} + 6\mathbf{k}|$.

2 The points A and B have coordinates $(1, 2, -1)$ and $(3, -4, 2)$. Find the distance AB.

3 **(Tricky)** Given $\mathbf{a} = \begin{pmatrix} t \\ 2 \\ t-1 \end{pmatrix}$ and $|\mathbf{a}| = 3$, find the possible values of t.

For the following statements, state whether it is true or false.

(a) $|\mathbf{i} + 2\mathbf{j} - 2\mathbf{k}| = 1 + 2 - 2 = 1$

(b) $|-\mathbf{a}| = |\mathbf{a}|$ for every vector $\mathbf{a}$

(c) $|\mathbf{a} + \mathbf{b}| = |\mathbf{a}| + |\mathbf{b}|$ for all vectors $\mathbf{a}$, $\mathbf{b}$

(d) $|3\mathbf{a}| = 3|\mathbf{a}|$

(e) $\left| \begin{pmatrix} 0 \\ -5 \\ 0 \end{pmatrix} \right| = -5$

(f) $\left| \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right| = 1$

(g) $|\mathbf{a} + \mathbf{b}| \leq |\mathbf{a}| + |\mathbf{b}|$ for all vectors $\mathbf{a}$, $\mathbf{b}$

3 Unit Vectors

1 Find the unit vector in the direction of $\begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}$.

2 Find the vector of magnitude 10 in the direction of $\begin{pmatrix} 3 \\ 0 \\ -4 \end{pmatrix}$.

3 **(Tricky)** The points A and B have coordinates $(1, 0, 2)$ and $(3, -6, 5)$. Find the vector of magnitude 14 in the direction *opposite* to $\overrightarrow{AB}$.

For the following statements, state whether it is true or false.

(a) $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ is a unit vector

(b) $\frac{1}{3}\begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$ is a unit vector

(c) $\mathbf{i} + \mathbf{j}$ is a unit vector

(d) $|\hat{\mathbf{a}}| = 1$ for every non-zero vector $\mathbf{a}$

(e) $\mathbf{a}$ and $\hat{\mathbf{a}}$ are parallel

4 Parallel Vectors ($\mathbf{a} = \lambda\mathbf{b}$)

1 Show that $\mathbf{a} = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} -6 \\ 3 \\ -9 \end{pmatrix}$ are parallel.

2 Find p and q such that $\begin{pmatrix} 4 \\ p \\ -2 \end{pmatrix}$ is parallel to $\begin{pmatrix} -2 \\ 3 \\ q \end{pmatrix}$.

3 **(Tricky)** Find all vectors of magnitude 6 that are parallel to $\mathbf{a} = \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}$.

For the following statements, state whether it is true or false.

(a) If $\mathbf{a} \parallel \mathbf{b}$, then $\mathbf{a} = \mathbf{b}$

(b) $\begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix}$ is parallel to $\begin{pmatrix} -2 \\ 4 \\ -6 \end{pmatrix}$

(c) $\begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix}$ is parallel to $\begin{pmatrix} 1 \\ 2 \\ 4 \end{pmatrix}$

(d) $\begin{pmatrix} 3 \\ 0 \\ -6 \end{pmatrix}$ is parallel to $\begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix}$

(e) If $\mathbf{a} = -2\mathbf{b}$, then $\mathbf{a}$ and $\mathbf{b}$ are not parallel since they point in opposite directions

5 Collinear Points

1 Show that the points $A(1, 2, 3)$, $B(3, 5, 4)$ and $C(7, 11, 6)$ are collinear.

2 For the points in Question 1, find the ratio $AB : BC$.

3 **(Tricky)** The points $P(2, -1, 0)$, $Q(4, 1, 3)$ and $R(a, 7, b)$ are collinear. Find a and b.

For the following statements, state whether it is true or false.

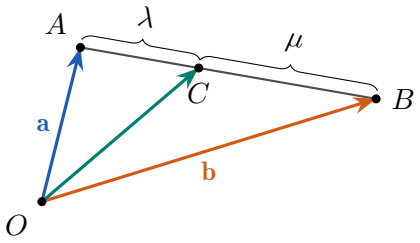
(a) If $\overrightarrow{AB} \parallel \overrightarrow{CD}$, then A, B, C and D are collinear

(b) If $\overrightarrow{AB} = 3\overrightarrow{AC}$, then A, B and C are collinear

(c) $A(0, 0, 0)$, $B(1, 1, 1)$ and $C(2, 2, 3)$ are collinear

(d) The points with position vectors $\mathbf{a}$, $2\mathbf{a}$ and $-\mathbf{a}$ are collinear ($\mathbf{a} \neq \mathbf{0}$)

6 Ratio Theorem



In the True/False statements, C lies on AB with $AC : CB = \lambda : \mu$, $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$.

1 Let $\overrightarrow{OA} = \mathbf{i} + \mathbf{j} + \mathbf{k}$ and $\overrightarrow{OB} = \mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$. Given that C lies on AB, find the coordinates of C if

(a) $AC : CB = 1 : 2$,

(b) $3AC = 2CB$,

(c) $2AB = 5CB$,

(d) Given instead that C lies on AB produced, and $3AC = 7BC$.

2 Find the position vector of the midpoint M of AB, where A and B have coordinates $(3, -1, 5)$ and $(-1, 5, 1)$.

For the following statements, state whether it is true or false.

(a) $\overrightarrow{OC} = \dfrac{\mu\mathbf{a} + \lambda\mathbf{b}}{\lambda + \mu}$

(b) $\overrightarrow{OC} = \dfrac{\lambda\mathbf{a} + \mu\mathbf{b}}{\lambda + \mu}$

(c) If $\lambda : \mu = 1 : 1$, then $\overrightarrow{OC} = \dfrac{1}{2}(\mathbf{a} + \mathbf{b})$

(d) If $\lambda : \mu = 1 : 3$, then $\overrightarrow{OC} = \dfrac{3\mathbf{a} + \mathbf{b}}{4}$

(e) If $\lambda : \mu = 2 : 1$, then $\overrightarrow{OC} = \dfrac{2\mathbf{a} + \mathbf{b}}{3}$

7 Scalar Product

1 It is given that $\mathbf{a} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} -3 \\ 2 \\ 7 \end{pmatrix}$, $\mathbf{c} = \begin{pmatrix} 4 \\ -5 \\ -1 \end{pmatrix}$. Find the following

a) $\mathbf{a} \cdot \mathbf{b}$
b) $\mathbf{b} \cdot \mathbf{a}$
c) $\mathbf{a} \cdot (\mathbf{b} + \mathbf{c})$
d) $\mathbf{a} \cdot \mathbf{b} + \mathbf{a} \cdot \mathbf{c}$
e) $\mathbf{a} \cdot (2\mathbf{c})$
f) $(2\mathbf{a}) \cdot \mathbf{c}$
g) $\mathbf{a} \cdot \mathbf{a}$

2

1) Points O, A, B are such that $\overrightarrow{OA} = \mathbf{i} + 4\mathbf{j} - 3\mathbf{k}$ and $\overrightarrow{OB} = 5\mathbf{i} - \mathbf{j}$
   (a) Find the size of angle AOB
   (b) Find the size of angle OAB.
2) It is given that $\mathbf{c} = \begin{pmatrix} 2 \\ 3 \\ 6 \end{pmatrix}$, $\mathbf{d} = \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix}$ and $\mathbf{e} = \begin{pmatrix} 3 \\ 2 \\ -2 \end{pmatrix}$. Find the angle between the vectors $\mathbf{c}$ and
   (a) the vector $\mathbf{d}$
   (b) the vector $\mathbf{e}$
   (c) the positive y-axis
3) It is given that two vectors $\mathbf{a}$ and $\mathbf{b}$ satisfy the following relationship:
$$|\mathbf{a}||\mathbf{b}| = 6.$$
   Find the value of $\mathbf{a} \cdot \mathbf{b}$ for each of the following scenarios.
   (a) The angle between $\mathbf{a}$ and $\mathbf{b}$ is $60°$.
   (b) The angle between $\mathbf{a}$ and $\mathbf{b}$ is $120°$.
   (c) Vectors $\mathbf{a}$ and $\mathbf{b}$ are perpendicular.

3

1) Referred to the origin O, the position vectors of the points A and B are $\mathbf{i} - \mathbf{j} + 2\mathbf{k}$ and $2\mathbf{i} + 4\mathbf{j} + \mathbf{k}$ respectively. Show that OA is perpendicular to OB.
2) The position vectors $\mathbf{a}$ and $\mathbf{b}$ are given by
$$\mathbf{a} = 2p\mathbf{i} + 3p\mathbf{j} + 6p\mathbf{k} \quad \text{and} \quad \mathbf{b} = \mathbf{i} - 2\mathbf{j} + 2\mathbf{k}$$
   where $p > 0$. It is given that $|\mathbf{a}| = |\mathbf{b}|$.
   (i) Find the exact value of p.
   (ii) Show that the vectors $(\mathbf{a} + \mathbf{b})$ and $(\mathbf{a} - \mathbf{b})$ are perpendicular.

4

1) Given that $\mathbf{a} = 2\mathbf{i} + \mathbf{j} + 2\mathbf{k}$ and $\mathbf{b} = -\mathbf{i} + 2\mathbf{k} - \mathbf{k}$, find the length of projection and the projection vector of
   (a) $\mathbf{a}$ onto $\mathbf{b}$,
   (b) $\mathbf{b}$ onto $\mathbf{a}$.
2) With reference to the origin O, the points A, B and C have position vectors given by
$$\mathbf{a} = \mathbf{i} + 2\mathbf{j} - \mathbf{k}, \ \mathbf{b} = 3\mathbf{i} + \mathbf{k} \text{ and } \mathbf{c} = 2\mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$$
   respectively. Find the length of projection of $\overrightarrow{BA}$ onto $\overrightarrow{BC}$.

For the following statements, state whether it is true or false.

(a) (Cancellation law) $\mathbf{a} \cdot \mathbf{b} = \mathbf{a} \cdot \mathbf{c} \implies \mathbf{b} = \mathbf{c}$

(b) $\mathbf{a} \cdot \mathbf{b} = \mathbf{c} \cdot \mathbf{d} \implies \mathbf{a} = \dfrac{\mathbf{c} \cdot \mathbf{d}}{\mathbf{b}}$

(c) $\mathbf{a} \cdot \mathbf{b} = 0 \implies \mathbf{a} = \mathbf{0}$ or $\mathbf{b} = \mathbf{0}$

8 Vector Product

1

1) Find $\begin{pmatrix} 3 \\ -1 \\ 4 \end{pmatrix} \times \begin{pmatrix} 5 \\ 7 \\ 1 \end{pmatrix}$

2) Given that $\mathbf{a} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ and $\mathbf{c} = \begin{pmatrix} -1 \\ 1 \\ -1 \end{pmatrix}$, find

(a) $\mathbf{a} \times \mathbf{b}$; (c) $\mathbf{a} \times (\mathbf{b} + \mathbf{c})$; (e) $\mathbf{a} \times (2\mathbf{b})$; (g) $\mathbf{a} \times \mathbf{a}$;
(b) $\mathbf{b} \times \mathbf{a}$; (d) $\mathbf{a} \times \mathbf{b} + \mathbf{a} \times \mathbf{c}$; (f) $(2\mathbf{a}) \times \mathbf{b}$;

2

1. Three non-zero vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ satisfy $\mathbf{a} \times 3\mathbf{b} = 2\mathbf{a} \times \mathbf{c}$. Show that $\mathbf{a}$ and $3\mathbf{b} - 2\mathbf{c}$ are parallel.
2. Find the scalar product and vector product of the following vectors, stating whether they are parallel, perpendicular, or neither.
   (i) $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ and $\begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}$
   (ii) $\mathbf{a} = \mathbf{i} + 2\mathbf{j} - 3\mathbf{k}$, $\mathbf{b} = 3\mathbf{i} + \mathbf{k}$
   (iii) $\begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$ and $\begin{pmatrix} -4 \\ -8 \\ 4 \end{pmatrix}$

3 Using the Vector Product, find the acute angles between the following vectors:

a) $\begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}$ and $\begin{pmatrix} 4 \\ -1 \\ 1 \end{pmatrix}$ b) $\begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix}$ and $\begin{pmatrix} 3 \\ -1 \\ 5 \end{pmatrix}$ c) $\begin{pmatrix} 4 \\ 3 \\ 5 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ -5 \\ 1 \end{pmatrix}$

4

1) Relative to an origin O, two point A and B have position vectors given by $\mathbf{a} = 3\mathbf{i} - \mathbf{j} + 2\mathbf{k}$ and $\mathbf{b} = -2\mathbf{i} + 3\mathbf{j} + 2\mathbf{k}$. Find the following:
   a) The shortest distance from A to OB
   b) The shortest distance from B to OA
2) Relative to the origin O, two points A and B have position vectors given by $\mathbf{a} = \mathbf{i} + \mathbf{j} + \mathbf{k}$ and $\mathbf{b} = \mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$ respectively.
   (i) Find the length of projection of $\mathbf{b}$ onto $\mathbf{a}$.
   (ii) Hence, or otherwise, find the shortest distance from B to OA.
3) Relative to the origin O, the points A, B and C have position vectors given by $\mathbf{a} = \mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$, $\mathbf{b} = \mathbf{i} - \mathbf{j} + \mathbf{k}$ and $\mathbf{c} = 3\mathbf{i} - \mathbf{j} + \mathbf{k}$. Find the shortest distance from C to the line joining A and B.
4) With respect to an origin O, the points A and B have position vectors $\mathbf{i} + 3\mathbf{j} - 4\mathbf{k}$ and $5\mathbf{i} + 2\mathbf{j} - 3\mathbf{k}$ respectively. F is the foot of perpendicular of point B on the line that passes through A and is parallel to the vector $\mathbf{d} = 2\mathbf{i} + \mathbf{j} + 4\mathbf{k}$
   (i) Find the exact length of AF
   (ii) Find the exact length of BF
   (iii) Find the vectors $\overrightarrow{AF}$ and $\overrightarrow{FB}$.

5

1) Relative to the origin O, the points A and B have position vectors $\overrightarrow{OA} = \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix}$, $\overrightarrow{OB} = \begin{pmatrix} -2 \\ 3 \\ 2 \end{pmatrix}$.
   Find the area of triangle OAB.
2) Relative to the origin O, the points A and B have position vectors $\overrightarrow{OA} = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$, $\overrightarrow{OB} = \begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix}$.
   Find the area of triangle OAB.
3) Relative to an origin O, the position vectors of the vertices A, B and C of a triangle are
   $$\mathbf{i} + \mathbf{j} - \mathbf{k}, \ 2\mathbf{i} + \mathbf{k} \text{ and } -\mathbf{i} + 2\mathbf{j} - \mathbf{k}$$
   respectively. By considering the displacement vectors $\overrightarrow{AB}$ and $\overrightarrow{AC}$, find
   (a) the exact area of triangle ABC, and
   (b) the exact length of the vector component of $\overrightarrow{AB}$ perpendicular to $\overrightarrow{AC}$.
4) The points A and B have coordinates $(5, 0, 5)$ and $(10, -5, 15)$ respectively. The point D is such that $OADB$ is a parallelogram and the point C divides AD in the ratio $2 : 3$. Find the exact areas of
   (i) the parallelogram $OADB$, and
   (ii) the quadrilateral $OCDB$.

END OF WORKSHEET